

PORTLAND BUOY 44007, ME, USA

WMO: 992780

Lat: 43.525N

Lon: 70.141W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (NAE)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	12.9	-4.3	4.4	10.0	-0.7	5.3	15.0	37.7	29.7	35.0	28.1	17.7	310	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	72.0	67.9	70.1	66.8	68.7	65.9	69.7	70.6	68.2	69.2	66.9	67.9	8.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	108.1	70.1	67.9	102.4	68.7	66.4	97.1	67.4	33.8	70.5	32.5	68.9	31.4	68.1	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.3	29.3	24.8	DB	3.0	77.3	4.8	3.6	-0.5	79.9	-3.3	82.0	-6.0	84.0	-9.5	86.5	(4)
(5)			WB	2.2	70.7	4.9	2.9	-1.3	72.7	-4.2	74.4	-6.9	76.0	-10.5	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	28.6	30.8	35.2	42.8	50.3	58.4	64.2	64.5	60.1	51.9	43.4	35.3
	DBStd	13.62	8.97	7.27	6.24	4.48	4.31	4.21	3.23	2.77	3.78	4.84	6.14	7.51
	HDD50	2634	665	539	460	220	49	0	0	0	0	36	209	456
	HDD65	6558	1129	959	924	667	455	200	55	42	152	405	649	921
	CDD50	1617	0	0	1	3	59	254	439	451	303	96	10	1
	CDD65	65	0	0	0	0	0	3	29	28	5	0	0	0
	CDH74	19	0	0	0	0	0	0	13	6	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.4	16.2	15.1	14.5	12.3	10.3	9.2	8.4	8.6	10.3	13.6	14.6	15.7
Precipitation	PrecAvg	49.1	3.3	3.4	4.2	4.2	3.8	4.5	3.7	3.5	4.2	5.3	4.4	4.6
	PrecMax	68.7	6.3	8.1	10.2	9.6	10.5	10.9	8.2	12.1	10.8	15.5	7.0	7.2
	PrecMin	33.6	0.3	1.6	1.0	0.2	0.9	1.1	1.5	0.5	1.2	1.0	1.3	1.7
	PrecStd	7.8	1.3	1.4	2.0	2.2	2.1	2.4	1.6	2.3	2.4	3.3	1.5	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	45.6	52.6	57.9	65.7	71.6	75.5	74.4	71.5	65.3	57.1	52.9
		MCWB	48.1	43.1	47.9	50.7	57.3	62.5	68.8	71.1	67.2	59.9	54.4	49.8
	2%	DB	46.3	43.5	46.9	52.2	60.9	67.9	72.4	71.6	68.4	61.7	54.8	49.9
		MCWB	44.0	41.5	43.4	48.4	56.1	62.1	67.9	68.5	65.1	58.6	51.8	47.5
	5%	DB	43.7	41.6	44.4	50.0	58.5	65.7	70.3	70.0	66.6	59.7	53.2	48.0
		MCWB	41.5	39.3	41.5	46.5	54.2	61.7	67.3	67.0	63.7	57.1	50.7	45.7
	10%	DB	40.7	40.0	42.3	48.2	56.2	64.0	68.7	68.7	65.3	58.2	51.8	45.7
		MCWB	37.9	37.5	39.7	45.3	52.6	61.1	66.5	66.0	62.7	55.7	49.4	43.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.7	44.8	50.2	53.3	61.1	66.6	71.6	73.2	70.7	62.3	56.2	51.2
		MCDB	49.1	45.3	50.4	55.5	62.2	68.0	72.8	74.2	71.2	63.4	57.1	52.1
	2%	WB	45.0	42.0	44.8	50.4	57.6	64.4	69.2	70.5	66.8	59.5	53.5	48.4
		MCDB	45.8	42.5	46.0	51.5	59.6	65.7	70.1	71.3	67.8	60.2	54.3	49.4
	5%	WB	41.6	40.1	42.4	48.1	55.1	63.1	68.0	68.5	65.3	58.1	51.7	46.4
		MCDB	43.3	41.2	43.8	49.3	56.7	64.6	69.1	69.5	66.4	59.0	52.6	47.7
	10%	WB	38.3	38.1	40.6	46.4	53.1	61.9	66.8	67.2	63.7	56.4	50.2	43.8
		MCDB	40.7	39.8	42.2	48.0	54.7	63.5	68.0	68.1	64.9	57.6	51.6	45.7
Mean Daily Temperature Range	5% DB	MDBR	10.6	9.7	8.2	6.7	6.2	6.2	6.4	6.1	6.0	6.9	7.6	9.6
		MCDBR	11.3	9.4	9.2	9.4	10.6	9.8	9.0	7.9	7.0	7.4	7.6	10.8
		MCWBR	11.8	10.0	8.0	7.3	7.5	6.0	5.8	6.2	6.0	6.8	8.3	10.2
	5% WB	MCDBR	11.0	8.9	8.6	8.6	9.0	8.1	7.4	7.6	6.5	6.3	7.1	10.0
		MCWBR	12.1	10.3	8.5	7.9	7.3	6.1	5.6	6.4	6.1	7.0	7.8	10.8
Clear-Sky Solar Irradiance	taub	0.298	0.308	0.331	0.358	0.392	0.444	0.457	0.425	0.380	0.341	0.321	0.299	
	taud	2.473	2.458	2.452	2.420	2.370	2.253	2.230	2.308	2.410	2.512	2.531	2.528	
	Ebn at Noon	266	283	289	288	279	264	259	264	268	267	256	256	
	Edh at Noon	23	27	31	35	38	43	43	39	32	26	22	20	
All-Sky Solar Radiation	RadAvg	527	810	1168	1474	1654	1770	1870	1697	1361	882	570	424	
	RadStd	35	60	73	154	186	146	122	84	77	79	52	47	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+142	N/A

Nomenclature: See separate page